Insurance Contact Centre Performance Analysis Using Amazon Connect Data

1. Project Overview

Objective

Analyse Amazon Connect contact centre data to identify causes of abandoned and missed calls and recommend operational improvements.

Tools Used

- PostgreSQL
- SQL
- Amazon Connect concepts
- Data Cleaning
- Business Analysis

Dataset

- 999 contact centre interactions
- 50 agents
- 3 queues: Claims, Sales, Renewals

2. Business Scenario

An insurance company uses Amazon Connect to manage customer interactions. Management has identified operational challenges impacting customer experience, including high abandoned call rates, increased missed calls, long wait times, and inconsistent agent performance. This analysis aims to identify the root causes of these issues and provide data-driven recommendations for improvement.

3. Business Questions

Customer Experience

- What is the abandoned call rate?
- What percentage of calls were missed?
- Which hours have the highest abandonment rate?
- What is the average wait time before abandonment?

Agent Performance

- Which agents handle the most calls?
- Which agents have the longest average handling time?

- Which agents have the most missed calls?
- Which agents resolve calls quickest?

Operational Analysis

- Are abandoned calls higher during peak periods?
- Is there enough staffing during busy hours?
- Which days experience the highest call volumes?

4. Data preparation

To ensure the reliability and accuracy of the analysis, a series of data quality assessments and cleansing activities were performed before conducting the detailed investigation.

Data Quality Assessment

The dataset was reviewed to identify inconsistencies, missing values, and duplicate records that could affect analysis accuracy and resulting business insights.

Call Status Standardisation

An assessment of call status values identified inconsistencies caused by data entry errors. Two call records contained the status value "Answerd" and one record contained "Missd", resulting in duplicate representations of valid statuses. These values were standardised to "Answered" and "Missed" using SQL update statements to ensure accurate reporting and aggregation of call outcomes.

Missing Agent Information

A review of agent-related data identified four records with missing agent names, represented by either NULL values or blank fields. These records were updated using the appropriate agent information to improve the completeness and reliability of agent performance analysis.

Duplicate Records

Data validation also identified a duplicate interaction record, where Call ID 1009 appeared more than once in the dataset. The duplicate entry was removed to prevent double-counting and ensure accurate calculation of contact centre performance metrics.

Outcome

Following the data cleansing process, the dataset was validated and deemed suitable for analysis. Addressing data quality issues ensured that subsequent findings, performance metrics, and business recommendations were based on accurate and reliable information.

5. Analysis & Findings

Key Findings

Analysis of 999 customer interactions showed that most calls were answered; however, many customers could not connect with an agent due to abandoned and missed calls.

Metrics	**Results**
Total calls	999
Answered calls	734
Abandoned calls	180
Missed calls	85
Abandoned rate	18%
Missed call rate	8.5%

Findings Summary

- Of the 999 customer interactions analysed, 73.5% of calls were answered, indicating the contact centre handled most customer requests.
- 180 calls were abandoned, resulting in an abandonment rate of 18%. This suggests that a notable number of customers ended the interaction before reaching an agent, potentially due to long wait times or insufficient staffing during peak periods.
- 85 calls were missed, representing 8.5% of total interactions. Missed calls may indicate operational inefficiencies, resource constraints, or call routing challenges.
- Combined, 26.5% of customer interactions were not successfully handled, highlighting opportunities to improve customer experience and operational performance.
- The findings indicate the need for further investigation into queue performance, agent availability, and peak demand periods to identify the key drivers of customer attrition.

Business Impact

The volume of abandoned and missed calls may contribute to:

- Reduced customer satisfaction.
- Increased customer effort and repeat call attempts.
- Potential loss of sales or policy retention opportunities.

- Higher operational costs due to repeat contacts.
- Increased strain on contact centre resources.

6. Recommendations

Increase Staffing During Peak Queue Periods

Analysis indicated that the Claims and Renewals queues experienced the highest volumes of abandoned calls. Workforce planning should be reviewed to ensure adequate agent coverage during peak demand periods to reduce customer wait times and improve service levels.

Monitor High Missed-Call Agents

Several agents consistently recorded higher missed-call volumes than their peers. A review of workloads, adherence metrics, network issues, and agent processes should be conducted to identify potential causes and provide targeted coaching where required.

Implement Real-Time Queue Monitoring

Leverage Amazon Connect dashboards and automated alerts to proactively monitor queue growth, service levels, and abandonment rates. Early identification of queue congestion can help supervisors take corrective action before customer experience is impacted.

Introduce Callback Functionality

Customers waiting beyond predefined thresholds should be offered a callback option. This can reduce abandonment rates, improve customer satisfaction, and distribute call volumes more effectively.

Review Workforce Planning and Scheduling

Call volume patterns should be analysed on an ongoing basis to align staffing levels with forecasted demand. Optimising schedules can improve resource utilisation while reducing abandoned and missed calls.

Establish Key Performance Indicator (KPI) Monitoring

Management should track critical contact centre metrics, including:

- Abandonment Rate
- Missed Call Rate
- Average Wait Time
- Average Handle Time
- Service Level Performance

Regular KPI monitoring will support data-driven decision-making and enable early identification of operational risks.

Investigate Root Causes of Long Wait Times

Further analysis should be conducted to understand factors contributing to extended customer wait times, including staffing availability, call routing efficiency, and queue management processes.

Provide Ongoing Agent Performance Reviews

Implement regular performance reviews using operational metrics to identify development opportunities, share best practices, and improve overall service quality across the contact centre.

7. Expected Business Impact

Implementation of the recommended improvements may result in:

- Reduced call abandonment and missed call rates.
- Improved customer satisfaction and service experience.
- More efficient allocation of contact centre resources.
- Better visibility into operational performance through KPI monitoring.
- Increased first-contact resolution opportunities.
- Lower risk of lost sales and customer retention issues caused by long wait times.